
THE REMIX

# Staggered DiD: Bacon + Callaway-Sant'Anna

*Day 3 of 3: Morning*

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## Goodman-Bacon (2021): the headline TWFE is not the ATT

**The question.** Did no-fault unilateral divorce laws reduce female suicide rates? Treatment timing varies across states from 1969 to 1985.

**The setup.** Staggered adoption + two-way fixed effects:

$$y_{ist} = \alpha_i + \gamma_t + \delta D_{ist} + \varepsilon_{ist}$$

**The headline result.** Stevenson-Wolfers report a meaningful negative effect: unilateral divorce reduced female suicide.

**Goodman-Bacon (2021) revisited it.** That headline TWFE coefficient is **not** the ATT. It's a weighted average of every 2×2 DiD you could draw with the data — including 2×2s where one already-treated state is used as the control for a later-treated state.

*Goodman-Bacon, A. (2021), "Difference-in-differences with variation in treatment timing," J. Econometrics 225(2): 254–277.*

## A small simulation makes the problem visible

**baker.do.** 1,000 firms, 4 cohorts of states, treated at 1986/1992/1998/2004. *Dynamic* treatment effects: ATT grows each year since treatment.

| | **The truth** | **TWFE estimate** |
|---|---|---|
| Mean ATT across cohorts and post-periods | $+82$ | $-6.69$ |

**The sign flipped.**

- Same data. Same DGP. Same TWFE regression you've been running.
- No econometric error in the regression itself — OLS gave you exactly what OLS gives.
- Something is wrong with the *estimand* TWFE is targeting.

*Simulation:* `the-remix-tour/simulations/staggered/baker.do`*; seed 20200403.*

# This morning, two acts

1. **Why TWFE goes wrong (Goodman-Bacon 2021).** TWFE is a variance-weighted average of every 2×2 DiD in the data. Some of those 2×2s use *already-treated* units as controls. When effects are dynamic, that's poison.

2. **Callaway & Sant'Anna (2021), the fix.** Forward-engineer from the target parameter ATT$(g, t)$. Estimate each one with the never-treated or not-yet-treated — never the already-treated. Then aggregate.

*The afternoon-morning thread: yesterday's DR-DiD is the engine inside CS's ATT(g,t).*

Lesson

# 1. Bacon's decomposition

TWFE is a weighted average of 2×2 DiDs — and OLS picks the weights

## Bacon's claim: TWFE *is* a weighted average of 2×2s

The TWFE regression with $K$ timing groups produces an estimate $\widehat{\delta}^{\text{TWFE}}$ that is *numerically identical* to a weighted average of every 2×2 DiD you can construct from the data.

### The decomposition (informally)

$$\widehat{\delta}^{\text{TWFE}} = \sum_{\text{every } 2\times 2 \text{ in the data}} s_{ij}\, \widehat{\delta}_{ij}^{2\times 2}$$

- This is *pure* OLS algebra. No assumptions about treatment effects yet.
- It's also not surprising: any OLS coefficient is a weighted average of some underlying "comparison" — Angrist (1998), Tymon (2022) for the general result.
- The interesting questions: **which 2×2s, with what weights, and is the average we get the average we want?**

## With $K$ timing groups, there are many 2×2s

Three timing groups $\{a, b, c\}$ plus one never-treated group $U$. Bacon counts the 2×2s:

| Treated vs. Never | Treated vs. Later | Treated vs. Earlier |
|---|---|---|
| $a$ vs. $U$ | $a$ vs. $b$ (later) | $b$ vs. $a$ (earlier) |
| $b$ vs. $U$ | $a$ vs. $c$ (later) | $c$ vs. $a$ (earlier) |
| $c$ vs. $U$ | $b$ vs. $c$ (later) | $c$ vs. $b$ (earlier) |

- Column 1: the comparisons you intended.
- Column 2: still uses an *eventually*-treated unit as a control — it's fine, because that unit hasn't been treated yet in the comparison window.
- Column 3: uses an *already*-treated unit as a control. **This is the forbidden comparison.**

## Two timing groups: $k$ (early) and $\ell$ (late)

Three time windows: PRE, MID (between the two treatment dates), POST.

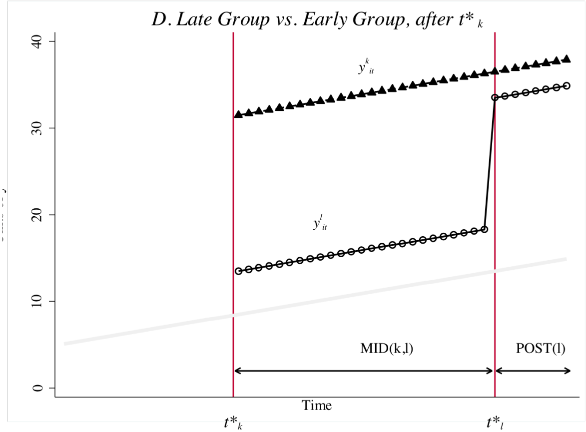
- **$k$ vs. $\ell$ during MID.** Group $k$ is post-treatment; group $\ell$ is still pre-treatment. **Comparison is fine** — $\ell$ is a not-yet-treated control.

$$\widehat{\delta}_{k\ell}^{2\times 2,\,k} = \left(\bar{y}_k^{\text{MID}} - \bar{y}_k^{\text{PRE}}\right) - \left(\bar{y}_\ell^{\text{MID}} - \bar{y}_\ell^{\text{PRE}}\right)$$

- **$\ell$ vs. $k$ during POST.** Group $\ell$ is post-treatment; group $k$ is *also* post-treatment. **The forbidden comparison** — $k$ is now an already-treated control.

$$\widehat{\delta}_{\ell k}^{2\times 2,\,\ell} = \left(\bar{y}_\ell^{\text{POST}} - \bar{y}_\ell^{\text{MID}}\right) - \left(\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}}\right)$$

# The forbidden 2×2, visualized




*Group $\ell$ treated in* POST*; group $k$ already treated since* MID*. One side is a treated unit that should never have been a control.*

# The full Bacon decomposition

**The setup.** $K$ timing groups, possibly one never-treated group $U$. TWFE estimate:

$$\widehat{\delta}^{\text{TWFE}} = \underbrace{\sum_{k \neq U} s_{kU}\, \widehat{\delta}^{2\times 2}_{kU}}_{\text{treated vs. never-treated}} + \underbrace{\sum_{k \neq U} \sum_{\ell > k} s_{k\ell} \Big[ \mu_{k\ell}\, \widehat{\delta}^{2\times 2,\,k}_{k\ell} + (1-\mu_{k\ell})\, \widehat{\delta}^{2\times 2,\,\ell}_{\ell k} \Big]}_{\text{treated vs. another-timing-group}}$$

- First sum: every timing group compared to the never-treated. Clean.
- Second sum: every *pair* of timing groups $(k, \ell)$ with $k$ earlier, $\ell$ later, contributes two 2×2s — one with $\ell$ as not-yet-treated control (fine), one with $k$ as already-treated control (**forbidden**).
- Weights $s_{kU}, s_{k\ell}$, and the within-pair weight $\mu_{k\ell}$ come from OLS algebra, not theory.

*Notation: we write the forbidden 2×2 as $\widehat{\delta}^{2\times 2,\ell}_{\ell k}$. Bacon (2021, Eq. 10a) writes it as $\widehat{\beta}^{2\times 2,\ell}_{k\ell}$. Same object; we follow Scott's CI-2 convention.*

## Where the weights come from: variance of $\tilde{D}$

After de-meaning $D_{it}$ within unit and time, what remains is $\tilde{D}_{it}$. The Bacon weights are functions of $\widehat{\mathrm{Var}}(\tilde{D}_{it})$ and group shares:

$$
\begin{aligned}
s_{kU} &= \frac{n_k\, n_U\, \overline{D}_k(1-\overline{D}_k)}{\widehat{\mathrm{Var}}(\tilde{D}_{it})} \\
s_{k\ell} &= \frac{n_k\, n_\ell\, (\overline{D}_k-\overline{D}_\ell)(1-(\overline{D}_k-\overline{D}_\ell))}{\widehat{\mathrm{Var}}(\tilde{D}_{it})} \\
\mu_{k\ell} &= \frac{1-\overline{D}_k}{1-(\overline{D}_k-\overline{D}_\ell)}
\end{aligned}
$$

- $\overline{D}_k$ is the share of group $k$'s panel spent in treatment.
- Same variance-weighting Angrist (1998) and Tymon (2022) describe generally: OLS weights treatment contrasts by treatment-status variance.

## Variance is maximized in the middle of the panel

Consider the never-treated 2×2 weight, $s_{kU}$. The key factor: $\overline{D}_k(1-\overline{D}_k)$.

| $\overline{D}_k$ (share of panel treated) | $\overline{D}_k(1-\overline{D}_k)$ |
|---|---|
| $0.1$ | $0.09$ |
| $0.3$ | $0.21$ |
| $0.5$ | $\mathbf{0.25}$ $\leftarrow$ **max** |
| $0.7$ | $0.21$ |
| $0.9$ | $0.09$ |

- Groups treated in the *middle* of the panel get the highest weight.
- Same logic for $(\overline{D}_k - \overline{D}_\ell)\big(1-(\overline{D}_k - \overline{D}_\ell)\big)$.
- **Panel length affects the weights.** Trim the panel and you've changed the estimand.

## So far: pure algebra. Now: potential outcomes.

What we have:

- TWFE is a weighted average of $2\times2$s.
- Some of those $2\times2$s use already-treated units as controls.
- OLS picks the weights based on treatment variance.

What we don't yet have:

- Whether the weighted average we get *is* the weighted ATT we want.

**To answer that we need to switch from realized outcomes to potential outcomes.**

*The next four slides walk through a single $2\times2$ step by step, then substitute back into the decomposition. The result is a clean expression for the bias.*

LESSON

# 2. The forbidden $2 \times 2$, step by step

Add a zero, rearrange, and the bias appears

## Step 1 — write down the 2×2 in observed outcomes

The 2×2 we're worried about: group $\ell$ (later-treated) in Post, with group $k$ (earlier-treated) as control.

$$\boxed{\widehat{\delta}_{\ell k}^{2\times 2,\,\ell} = \left[\bar{y}_\ell^{\text{POST}} - \bar{y}_\ell^{\text{MID}}\right] - \left[\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}}\right]}$$

- Both groups are *treated* in Post.
- Both groups were *treated* in Mid? No — $\ell$ is still untreated in Mid.
- So group $\ell$'s "change" includes its move from untreated to treated. Group $k$'s "change" is from *treated* to *treated*.

## Step 2 — switch to potential outcomes

Replace each observed mean with the appropriate potential-outcome mean (using the switching equation):

$$
\begin{aligned}
\bar{y}_\ell^{\textsc{Post}} &= \mathbb{E}[Y^1 \mid \ell, \textsc{Post}] && \text{(treated)} \\
\bar{y}_\ell^{\textsc{Mid}} &= \mathbb{E}[Y^0 \mid \ell, \textsc{Mid}] && \text{(not yet treated)} \\
\bar{y}_k^{\textsc{Post}} &= \mathbb{E}[Y^1 \mid k, \textsc{Post}] && \text{(treated)} \\
\bar{y}_k^{\textsc{Mid}} &= \mathbb{E}[Y^1 \mid k, \textsc{Mid}] && \text{(already treated)}
\end{aligned}
$$

**Notice the asymmetry.** Group $k$ is in $Y^1$ in *both* MID and POST. That's the problem.

*Three of the four cells are $Y^1$. Only one is $Y^0$.*

## Step 3 — add zeros

We add three zero-valued terms of the form $\mathbb{E}[Y^0 \mid \cdot] - \mathbb{E}[Y^0 \mid \cdot]$:

### The three zeros

1. $+\mathbb{E}[Y^0 \mid \ell, \textsc{Post}] - \mathbb{E}[Y^0 \mid \ell, \textsc{Post}]$ (add to $\ell$ in Post)
2. $+\mathbb{E}[Y^0 \mid k, \textsc{Post}] - \mathbb{E}[Y^0 \mid k, \textsc{Post}]$ (add to $k$ in Post)
3. $+\mathbb{E}[Y^0 \mid k, \textsc{Mid}] - \mathbb{E}[Y^0 \mid k, \textsc{Mid}]$ (add to $k$ in Mid)

- Each adds zero. The 2×2 is unchanged.
- But now we can group the terms into something interpretable.

## Step 4 — rearrange into ATT, parallel-trends, and $\Delta$ATT

Group the eight terms (four observed + four added zeros) into three buckets:

### The forbidden 2×2 in causal pieces

$$\widehat{\delta}^{2\times 2,\,\ell}_{\ell k} = \underbrace{\mathrm{ATT}_\ell(\textsc{Post})}_{\text{what we want}} + \underbrace{\Delta Y^0_\ell(\textsc{Post},\textsc{Mid}) - \Delta Y^0_k(\textsc{Post},\textsc{Mid})}_{\text{parallel-trends bias}} - \underbrace{\left[\mathrm{ATT}_k(\textsc{Post}) - \mathrm{ATT}_k(\textsc{Mid})\right]}_{\text{heterogeneity bias: } \Delta\mathrm{ATT}_k}$$

- Bucket 1 is the ATT we want for group $\ell$ in the post-period.
- Bucket 2 is the standard PT bias — requires PT between $\ell$ and $k$ in this window.
- Bucket 3 is **new**: it's the *change* in group $k$'s own ATT between Mid and Post. If treatment effects grow (dynamics), this is positive — and it enters with a *minus sign*.

## Step 5 — substitute back into Bacon's decomposition

The three flavors of 2×2 in causal pieces:

$$
\begin{aligned}
\widehat{\delta}^{2\times 2}_{kU} &= \text{ATT}_k(\textsc{Post}) \;+\; \Delta Y^0_k(\text{post},\text{pre}) - \Delta Y^0_U(\text{post},\text{pre}) \\
\widehat{\delta}^{2\times 2,\,k}_{k\ell} &= \text{ATT}_k(\textsc{Mid}) \;+\; \Delta Y^0_k(\textsc{Mid},\text{pre}) - \Delta Y^0_\ell(\textsc{Mid},\text{pre}) \\
\widehat{\delta}^{2\times 2,\,\ell}_{\ell k} &= \text{ATT}_\ell(\textsc{Post}) \;+\; \Delta Y^0_\ell(\textsc{Post},\textsc{Mid}) - \Delta Y^0_k(\textsc{Post},\textsc{Mid}) \\
&\quad - \;\Delta\text{ATT}_k
\end{aligned}
$$

Plug these into Bacon's weighted sum. The TWFE probability limit:

$$
\textbf{plim}\,\widehat{\delta}^{\textbf{TWFE}} \;=\; \textbf{VWATT} \;+\; \textbf{VWPT} \;-\; \boldsymbol{\Delta}\textbf{ATT}
$$

*Bacon (2021, Eq. 15) writes VWCT (variance-weighted common trends). Same object as our VWPT.*

## What the decomposition tells us

$$\operatorname{plim} \widehat{\delta}^{\mathsf{TWFE}} = \mathsf{VWATT} + \mathsf{VWPT} - \Delta\mathsf{ATT}$$

- **VWATT** — a variance-weighted ATT. Not the ATT *you* want unless you specifically wanted variance-weighted averaging. The weights depend on *panel length* (a researcher choice).
- **VWPT** — variance-weighted parallel-trends bias. Zero only if PT holds across *every* pairwise comparison, including between treated groups.
- **$\Delta$ATT** — the dynamics term. If ATT grows with time-since-treatment, this is *positive* and enters with a *minus sign* — attenuating TWFE toward zero, or even **flipping the sign**.

*That's exactly what happens in baker.do: dynamics big enough that $-\Delta$ATT swamps the rest, flipping $+82$ to $-6.69$.*

Lesson

# 3. baker.do, in code

Set the seed, write the DGP, compute the truth, run TWFE, watch it fail

# The panel: 30 explicit lines, because repetition *is* the lesson

```
set obs 40
gen state = _n
expand 25
bysort state: gen firms = runiform(0,5)
expand 30
sort state
bysort state firms: gen year = _n

replace year = 1980 if year==1
replace year = 1981 if year==2
replace year = 1982 if year==3
replace year = 1983 if year==4
replace year = 1984 if year==5
replace year = 1985 if year==6
replace year = 1986 if year==7 // cohort 1 first treated here
replace year = 1987 if year==8
replace year = 1988 if year==9
... (22 more explicit lines, one per year, through 2009) ...
```

*A loop would be shorter. The 30 lines are the lesson: each is one substitution, one beat. See `simulations/README.md` §1.*

## The DGP, in 12 more explicit lines

```
set seed 20200403

* Group-specific TE means (heterogeneous across cohorts)
gen te1 = rnormal(10, 0.04) // cohort 1986
gen te2 = rnormal( 8, 0.04) // cohort 1992
gen te3 = rnormal( 6, 0.04) // cohort 1998
gen te4 = rnormal( 4, 0.04) // cohort 2004
gen te = cond(group==1, te1, cond(group==2, te2, cond(group==3, te3, te4)))

* Y(0) and Y(1) under DYNAMIC effects (te grows since treatment)
gen y0 = firms + n + e
gen te_dynamic = te*treat*(year - treat_date + 1)
gen y = y0 + te_dynamic
```

*One step per line, named variables, no loops. Full file:* `simulations/staggered/baker.do`*.*

# The truth, computed before running anything

```
* The truth (averaged across cohorts and post-periods)
su te_dynamic if treat == 1
display "True average ATT (dynamic) = " r(mean)
```

Output: `True average ATT (dynamic) = 82`

**Why 82?** The dynamic effect for group 1 averages $(10 + 20 + \cdots + 240)/24$. Combined across cohorts and post-periods, the simple average is 82.

*We computed the truth from the DGP. Now compare it to what TWFE produces.*

# TWFE, applied

```
areg y i.year treat, a(id) robust
```

| | Truth | TWFE |
|---|---|---|
| $\widehat{\text{ATT}}$ | $+82$ | $-6.69^{***}$ |

**Sign-flipped.** Extreme dynamics — the $-\Delta$ATT term swamps VWATT.

## The Bacon decomposition of that $-6.69$

| DD comparison | Weight | Avg DD estimate |
|---|---|---|
| Earlier-treated vs. later-treated-(as-control) | $0.500$ | $+51.80$ |
| Later-treated vs. earlier-treated-(as-control) | $0.500$ | $-65.18$ |

$$(0.5)(+51.80) \ + \ (0.5)(-65.18) \ = \ -6.69.$$

**Read it.** Half the weight is on a 2×2 averaging $-65$ — a 2×2 that uses an already-treated group as control. **That's the forbidden comparison doing the damage.**

## Bacon's own decomposition of Stevenson-Wolfers

We opened with Stevenson-Wolfers. Bacon ran the same decomposition on their actual data.

$$\widehat{\beta}^{\text{DD}}_{\text{SW}} = -3.08 \qquad \text{(156 components)}$$

- The late-vs-early-treated 2×2s average $+3.51$ — the wrong sign, pulled by treatment-effect heterogeneity over time.
- Drop those terms: the rest averages $-5.44$, close to the event-study average ($-4.92$).
- The single TWFE coefficient of $-3.08$ is masking a $+3.51$ vs. $-5.44$ split. **Same disease as baker.do, less extreme.**

*Goodman-Bacon (2021, §4 and Fig. 5). The SW headline survives qualitatively but its magnitude is contaminated.*

LESSON

# 4. Visualizing staggered structure

What the cells look like, what TWFE averages over

# Constant-TE worksheet

| Year | ATT(1986,t) | ATT(1992,t) | ATT(1998,t) | ATT(2004,t) |
|---|---|---|---|---|
| 1980–1985 | 0 | 0 | 0 | 0 |
| 1986–1991 | 10 | 0 | 0 | 0 |
| 1992–1997 | 10 | 8 | 0 | 0 |
| 1998–2003 | 10 | 8 | 6 | 0 |
| 2004–2009 | 10 | 8 | 6 | 4 |
| ATT$(g)$ | 10 | 8 | 6 | 4 |

- **Simple ATT** (uniform avg over colored cells) $= 8.0$.
- **VWATT** (TWFE's target) $\neq 8.0$ — OLS weights down edge cohorts via $\overline{D}(1 - \overline{D})$.

*Even with constant effects per cohort, TWFE doesn't give the simple ATT.*

## Dynamic-TE worksheet

| Year | ATT(1986,t) | ATT(1992,t) | ATT(1998,t) | ATT(2004,t) |
|---|---|---|---|---|
| 1986 | 10 | 0 | 0 | 0 |
| 1992 | 70 | 8 | 0 | 0 |
| 1998 | 130 | 56 | 6 | 0 |
| 2004 | 190 | 104 | 42 | 4 |
| 2009 | 240 | 144 | 72 | 24 |

ATT grows each year since treatment (the dynamic DGP). Simple ATT (uniform avg over colored cells) $\approx 82$; TWFE $= -6.69$.

**The $-\Delta$ATT bias is doing the damage.** Late cohorts use the already-treated earlier cohorts as “controls” — whose outcomes are growing.

LESSON

# 5. Callaway & Sant'Anna

Forward-engineer from ATT$(g, t)$, never use the already-treated as control

# The forward-engineering principle

Bacon told us what reverse-engineering from TWFE buys you: a variance-weighted average of every 2×2 in the data, including the forbidden ones.

**The Callaway-Sant'Anna (2021) inversion:**

1. **Start** with the target parameter you actually want: ATT$(g, t)$ — the average treatment effect on cohort $g$ in calendar period $t$.
2. **State** the identification assumptions formally.
3. **Derive** an estimator directly from those assumptions.
4. **Aggregate** the building blocks into whatever summary you want.

*Each ATT$(g, t)$ is its own clean 2×2 problem. Yesterday afternoon's DR-DiD is exactly the engine that estimates one.*

# ATT$(g, t)$ — the building block

**Definition**

$$\text{ATT}(g,t) \;=\; \mathbb{E}\big[\, Y_{i,t}(1) - Y_{i,t}(0) \,\big|\, G_i = g \,\big]$$

- $G_i = g \;=\;$ "unit $i$ was first treated in period $g$." (Cohort identifier.)
- One ATT$(g, t)$ for every (cohort, calendar-period) pair where the cohort is treated.
- In baker.do: 4 cohorts $\times$ (varying numbers of post-periods) $=$ 60 ATT$(g, t)$ cells in the grid.
- Each cell is a single 2$\times$2 estimand. **The aggregation is a separate, downstream choice.**

## Three flavors of parallel trends for staggered settings

The $2 \times 2$ PT assumption generalizes three ways:

| **Assumption** | **Comparison group** | **When to use** |
|---|---|---|
| PT-GT-Nev | Never-treated only | Conservative; needs a never-treated to exist |
| PT-GT-NYT | Not-yet-treated | **Authors' preferred** — middle ground |
| PT-GT-all | All groups, all periods | Strongest assumption; imposes parallel pre-trends |

- NYT lets late-treated groups serve as controls for earlier-treated groups in their pre-treatment window. Never as already-treated controls.
- The forbidden $2 \times 2$ of Bacon is *ruled out by construction*.

*Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025), "DiD: A Practitioner's Guide."*

## The estimator — and yesterday's DR is the recommended one

Three options for the building-block estimator of each ATT$(g,t)$:

1. **OR.** Heckman-Ichimura-Todd: model $\Delta Y(0)$ on $X$, impute the treated counterfactual.
2. **IPW.** Abadie (2005): weight controls by $p(X)/(1-p(X))$.
3. **DR.** Sant'Anna-Zhao (2020). Consistent if *either* model is correct.

**The practitioner's-guide recommendation: "Favor doubly robust."**

$$\widehat{\delta}_{\text{DR}}(g,t) \;=\; \mathbb{E}\Big[w_{\text{DR}}(X)\,\big(\Delta Y \;-\; \widehat{\mu}_0(X)\big)\Big]$$

**The DR-DiD you proved yesterday afternoon. Same engine, applied cell-by-cell to every feasible $(g,t)$.**

## Aggregation is a downstream choice, not an estimation choice

Once you have $\{\widehat{\text{ATT}}(g,t)\}$, build whatever summary the policy question demands:

- **Simple ATT.** Uniform average over all post-treatment $(g,t)$ cells.
- **Group ATT, ATT$(g)$.** Average each cohort's own post-treatment cells. "What did the policy do to cohort $g$ on average?"
- **Event-time ATT, ATT$_{\text{es}}(e)$.** Average across cohorts at fixed event time $e$ since treatment. "What's the dynamic profile?"
- **Calendar-time ATT.** Average across cohorts within a calendar period. "What happened in 2002, on average?"

**Inference.** Bootstrap, with uniform confidence bands when reporting an event-study (the `did` R package does this by default).

# baker_cs.do — the implementation

```
* Get the same baker data, fit CS with DR-DiD
ssc install csdid, replace
ssc install drdid, replace

csdid y, ivar(id) time(year) gvar(treat_date) ipw long2

* Four aggregations, four answers:
csdid_estat simple // simple ATT across all cells
csdid_estat group // ATT(g) per cohort
csdid_estat calendar // ATT by calendar period
csdid_estat event // event-time profile
```

*Building blocks first, aggregation second.*

*collegio-carlo-alberto/labs/Baker/Solutions/baker_cs.do.*

## The headline: CS recovers the truth

| | Truth | TWFE | CS (DR-DiD) |
|---|---|---|---|
| Feasible ATT (pre-2004 sample) | $68.33$ | $26.81^{***}$ | $\mathbf{68.34}^{***}$ |

- The “feasible” sample drops the 2004 cohort so a never-treated comparison exists.
- TWFE is no longer sign-flipped (no universally-treated tail), but it still attenuates — still has VWATT weights, still uses early-treated cohorts as controls.
- CS lands on the truth to two decimals.

## Not every ATT$(g, t)$ is feasible

CS can only estimate ATT$(g, t)$ if there is a valid comparison group for $(g, t)$:

- Under **PT-GT-Nev**: need a never-treated unit observed at $t$.
- Under **PT-GT-NYT**: need a not-yet-treated unit observed at $t$.

**In baker.do:** The last cohort is treated in 2004. After 2004, every unit is treated. **No comparison units exist for ATT(g, t > 2004).**

**Two responses:**

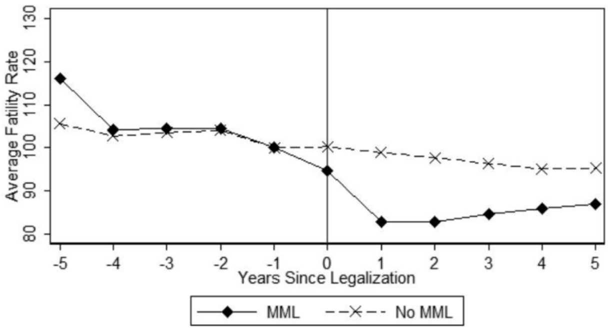
- Restrict to the feasible sample. CS will tell you which $(g, t)$ it dropped.
- Or, if you really want estimates for the universally-treated tail, use a different framework (Imputation: Borusyak et al. 2024; Wooldridge 2021) that doesn't require a comparison group.

Lesson

# 6. Sun & Abraham (2021)

Bacon was static. Event-studies have their own contamination.

# Researchers love event studies — and TWFE will lie to you on them




*Anderson et al. (2013): raw traffic-fatality rates for treated states (re-centered to treatment date) vs. controls. The classic dynamic-effects display.*

## The classic dynamic-TWFE event-study spec

Researchers want to plot pre-treatment placebo coefficients and post-treatment dynamic effects. The standard regression:

$$Y_{i,t} \;=\; \alpha_i \;+\; \delta_t \;+\; \sum_{g \in G} \mu_g \, \mathbf{1}\{t - E_i \in g\} \;+\; \varepsilon_{i,t}$$

- $E_i$ is unit $i$'s treatment date. $t - E_i$ is event time (relative-to-treatment).
- $G$ is the set of event-time bins (leads and lags) included.
- Coefficient $\mu_g$ is supposed to be the average treatment effect at event-time bin $g$.

**Bacon told us static TWFE is biased.** Sun & Abraham (2021) ask: *what about the dynamic version?*

## SA's proposition: $\mu_g$ is contaminated by other leads and lags

Even under parallel trends and no anticipation, the TWFE event-study coefficient on bin $g$ is:

$$\begin{aligned}
\mu_g \;=\; & \underbrace{\textstyle\sum_{l\in g}\sum_e w^g_{e,l}\,\text{CATT}_{e,l}}_{\text{the target (own bin)}} \\
& + \underbrace{\textstyle\sum_{g'\neq g}\sum_{l'\in g'}\sum_e w^g_{e,l'}\,\text{CATT}_{e,l'}}_{\text{contamination from \textit{other} included bins}} \\
& + \underbrace{\textstyle\sum_{l'\in g^{\text{excl}}}\sum_e w^g_{e,l'}\,\text{CATT}_{e,l'}}_{\text{contamination from \textit{excluded} bins}}
\end{aligned}$$

**Other-bin weights sum to zero — but only cancel under treatment-effect homogeneity.** With heterogeneous CATTs, the contamination doesn't cancel.

## The disturbing corollary: pre-period $\mu_g$ can be nonzero even when PT holds

Under PT $+$ no anticipation but *not* homogeneity:

- A pre-period lead $\mu_{-2}$ should be zero if pre-trends are truly parallel.
- In TWFE it isn't — it picks up CATTs from *post-treatment* bins through the cross-bin weights.
- So a "failed placebo" may just be the cross-bin contamination.
- And a "passing placebo" may just be cancellation, not evidence of PT.

**Lesson:** Don't drop excluded bins arbitrarily. Lagged effects contaminate through the excluded bins.

*Same disease as Bacon's static decomposition, but applied to leads and lags — so it affects the entire plot you're showing the audience.*

## SA's fix: the interaction-weighted estimator (3 steps)

1. **Estimate each $\widehat{\delta}_{e,l}$ separately.** A saturated regression with cohort $\times$ event-time interactions, using the never-treated (or last-treated) as control:

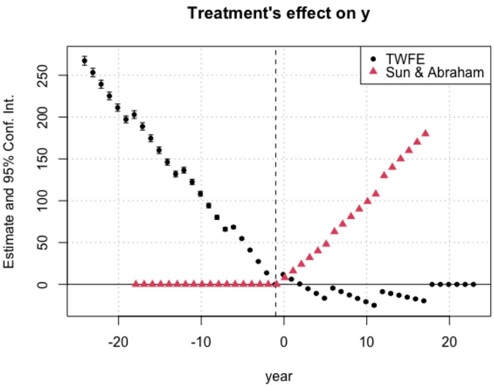
$$Y_{i,t} \;=\; \alpha_i \;+\; \lambda_t \;+\; \sum_{e \notin C} \sum_{l \neq -1} \delta_{e,l} \left[ \mathbf{1}\{E_i = e\} \cdot D^l_{i,t} \right] \;+\; \varepsilon_{i,t}$$

2. **Estimate cohort shares** in the relevant periods: $\widehat{\Pr}(E_i = e \mid E_i \in [-l, T-l])$.
3. **Aggregate** the $\widehat{\delta}_{e,l}$ using the cohort shares as weights.

**Same logic as CS.** Estimate cohort-event-time cells first; aggregate after.
**Forward-engineer the parameter.** *Software: `eventstudyinteract` (Stata, SSC);*

*`fixest::sunab` (R).*

# TWFE event study vs. SA, side by side




*TWFE leads slope downward pre-treatment — the cross-bin contamination. SA's pre-period sits at zero; lags trace the dynamic profile.*

LESSON

# 7. de Chaisemartin & D'Haultfœuille

Negative weights on *unit-level* effects, plus the on-off case

## dCdH (2020) adds three things to the staggered toolkit

1. **A different decomposition.** Bacon decomposed TWFE into $2\times2$ building blocks with *positive* weights. dCdH decomposes TWFE into *unit-level* treatment effects — where weights can be **negative**.
2. **The on-off case.** CS and SA assume irreversible treatment. dCdH handles units that switch into and out of treatment (think: a state policy that was repealed). Roe v. Wade adopted in 1973, overturned in 2022.
3. **A diagnostic.** Run the TWFE you were going to run, then count the *share of unit-period treatment effects* that receive negative weights. Report the sign-flip threshold.

## The dCdH decomposition: weights on unit-level effects

Let $\Delta^g_{i,t} = Y^1_{i,t} - Y^\infty_{i,t}$ be the *unit-level* treatment effect for unit $i$ in period $t$. Under PT, no anticipation, and SUTVA:

**Theorem (dCdH 2020)**

$$\delta^{\text{TWFE}} = \mathbb{E}\Big[\sum_{i,t:\,D_{i,t}=1} \frac{1}{NT}\, w_{i,t}\, \Delta^g_{i,t}\Big], \qquad \sum_{i,t:\,D_{i,t}=1} \frac{w_{i,t}}{NT} = 1, \quad w_{i,t}\ \textbf{can be negative}.$$

- Weights come from the OLS residual of $D_{i,t}$ on unit and time FE.
- Bacon's weights are on *2×2s* (and always positive). dCdH's weights are on *individual treatment effects* (and may be negative).

# Negative weights break the "no sign-flip property"

**Thought experiment.** Suppose every unit gains from the treatment ($\Delta^g_{i,t} > 0$ for all $i, t$).

- If all $w_{i,t} > 0$: $\widehat{\delta}^{\text{TWFE}}$ is a positive weighted average of positive effects. Sign matches.
- If some $w_{i,t} < 0$: nothing prevents $\widehat{\delta}^{\text{TWFE}}$ from being **negative** — even though every unit gained.

**This is what we saw in baker.do.** Truth $+82$, TWFE $-6.69$. Some unit-level effects got large negative weights; they swamped the positive-weight contributions. *Pre-dCdH, "OLS is a weighted average" was reassuring. dCdH made it disturbing.*

# When the negative weights bite (and when they don't)

**The diagnostic dCdH recommends running, before you publish anything:**

1. Fraction of unit-period treatment effects receiving *negative* weight in your TWFE.
2. The minimum amount of heterogeneity in $\Delta_{i,t}^{g}$ that would flip the sign of $\widehat{\delta}^{\mathrm{TWFE}}$.

**Rules of thumb:**

- Large never-treated group $\Rightarrow$ fewer negative weights, smaller magnitudes.
- Treatment dates packed close together $\Rightarrow$ also fewer.
- Long panel with a few early-treated cohorts $\Rightarrow$ the worst case.

*Software: `twowayfeweights` (Stata, SSC; R via the `TwoWayFEWeights` package) reports both diagnostics on your specific dataset.*

## The on-off case: `did_multiplegt` for switching treatment

Suppose treatment can switch on *and* off (Roe v. Wade 1973 $\rightarrow$ Dobbs 2022; minimum-wage rollbacks; tax cuts that sunset).

- CS and SA both assume **irreversible** treatment — they cannot estimate effects in this setting.
- dCdH propose a different estimator built on *instantaneous* effects between adjacent periods, comparing units that just switched against units whose treatment status is unchanged.
- Equivalent to CS & SA when treatment is irreversible (with their particular weighting), but *also* handles the on-off case.

*Software: `did_multiplegt` (Stata); the `DIDmultiplegt` R package.*

# The unifying lesson across Bacon, SA, and dCdH

## What makes weights "good"

The economic question is: *what parameter do you want?*

OLS doesn't ask. OLS picks weights from the residual variance of $D$.

- Bacon: TWFE = weighted average of 2×2s. Weights positive but variance-driven.
- SA: event-study $\mu_g$ = weighted average of $\text{CATT}_{e,l}$ *across* bins. Cross-bin contamination.
- dCdH: TWFE = weighted average of unit-level $\Delta^g_{i,t}$. **Weights can be negative.**

**All three say the same thing:** If you let OLS pick the weights, you've let it pick your economic question for you. Forward-engineer instead.

Lesson

# 8. Synthesis

What's the same, what's different, what to do tomorrow morning

# Yesterday afternoon plus this morning = one workflow

- **Yesterday (covariates).** Conditional PT $\Rightarrow$ DR-DiD. Two chances to be right (OR model or pscore model).
- **This morning (staggered).** ATT$(g,t)$ building blocks, estimated cell-by-cell, then aggregated.
- **The combination.** Each ATT$(g,t)$ *is* a 2×2 problem. The recommended way to estimate it is DR-DiD with whatever $X$ you need for conditional PT. Then aggregate.

## One sentence

**Forward-engineer the target, estimate the cells with DR, aggregate with non-negative weights.**

# Basically never.

- TWFE = ATT only when (a) treatment effects are homogeneous across cohorts, AND (b) treatment effects don't change over time within a cohort.
- Both fail in almost every applied setting.
- Even when they hold, the variance weights are still strange (depend on panel length).
- **If you must report it, also report CS, SA, or dCdH — and the dCdH negative-weight diagnostic. Explain the differences.**

*Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025): "Do not use TWFE."*

## A student asks: "But what about my paper from last year that used TWFE?"

A reasonable thing to do, in order:

1. **Run the Bacon decomposition** (`ddtiming`; `bacondecomp` in R). See whether the forbidden $2{\times}2$s carry weight.
2. **Run the dCdH diagnostic** (`twowayfeweights`). Sizeable negative-weight share $\Rightarrow$ headline is suspect.
3. **Re-estimate with CS or SA.** Compare the new headline to TWFE.
4. **If they disagree:** the difference *is* the $-\Delta$ATT / VWPT / negative-weight contamination. Trust the design-based estimator.
5. **If they agree:** in your setting, dynamics and heterogeneity didn't bite. Report both, explain.

*Goodman-Bacon, Sant'Anna, Whitten (2026) gives a refined re-analysis framework for pre-trends-parallel settings.*

## Lab: rerun baker, rerun baker_cs, compare

- **Step 1.** Run `simulations/staggered/baker.do` to (re)generate the dynamic-TE dataset. Confirm the truth is $+82$ and TWFE is $-6.69$.
- **Step 2.** Run the Bacon decomposition (`ddtiming`). Verify the two weights and the two $2\times2$ averages reproduce the textbook table.
- **Step 3.** Run `baker_cs.do`: CS with DR-DiD, plus four aggregations (`simple`, `group`, `calendar`, `event`).
- **Step 4.** Drop the 2004 cohort. Re-run TWFE: notice it stops sign-flipping (no longer have an universally-treated tail). Re-run CS: now lands on the truth.
- **Step 5.** Write a one-paragraph summary: which assumption is doing the work, and what would have to be true for TWFE to be correct?

*Walkthrough: `zurich/labs/Baker/README.md`. Code: `simulations/staggered/baker.do`, plus the CS files Scott will provide.*

## Tomorrow afternoon

- **Honest event-studies.** Roth (2022) on the perils of conditioning on pretest passes. Rambachan & Roth (2023) partial-identification bounds.
- **Imputation as a third path.** Borusyak-Jaravel-Spiess (2024) / Wooldridge (2021) extended TWFE / Gardner (2021): consistent estimators when no comparison group exists.
- **Continuous and multi-valued treatment.** Dose-specific ATT, average causal response (Callaway, Goodman-Bacon, Sant'Anna 2024).
- **Goodman-Bacon, Sant'Anna, Whitten (2026)** — a refined re-analysis framework for "pre-trends parallel" settings, with sharper conditions for when TWFE event-studies can be trusted.

Reading for tonight: *Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025)*
*"Difference-in-Differences Designs: A Practitioner's Guide."*

# Forward-engineer.

*Start with what you want to estimate.*

*Match the estimator to the assumption.*
*Aggregate at the end, not the start.*

Scott Cunningham · University of Zurich · May 13, 2026